Anirudh Srinivasan

SENIOR AI RESEARCH ENGINEER, BHARATGEN

anirudhsrinivasan.tms@gmail.com — srini-anirudh — srini-anirudh.github.io — anirudh-srini — Anirudh_Srinivasan3

Education

Indian Institute of Technology, Hyderabad *CGPA: 8.76/10.00*

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING : **INSTITUTE GOLD MEDAL FOR EXCELLENCE IN ACADEMICS AND CO-CURRICULAR ACTIVITIES (RANK 1 OUT OF 1103)** *July 2020 - May 2024*

Key Coursework: Data Structures, Algorithms, DBMS, Applied Statistics, Data Science Analytics, Convex Optimization, Markov Chains, Neuroscience & BCI, Foundations of ML, Deep Learning, Image Processing, Video Content Analysis, NLP, Tensors, Quantum Computing

Publications

C=Conference Paper, W=Workshop Paper, P=Preprint, S=Under Submission. * denotes Equal Contribution

[S.1] DocLayout-VL: A Foundational Model for Hierarchical, Open-set, and Promptable Document Layout Segmentation
Srinivasan, A.*, Venna, V. K.*, Bandarupalli, S.*, Raghuveer, R., Gunda, S. M., & Sarvadevabhatla, R. K. (under submission at **ECCV**)

[C.2] REPLICA: An Agentic Framework for Visually Faithful Document Reconstruction
Srinivasan, A.*, Raghuveer, R.*, Venna, V. K.*, Sreevatsa, T., Jain, A., Kukkala, S., & Sarvadevabhatla, R. K. (Poster at **ICDAR '26**)

[C.3] Patram-Bench: Comprehensive Multi-task, Multi-domain and Multilingual Benchmark for Indian Document Understanding
Srinivasan, A.*, Jena, P.*, Topale, A., Venna, V. K., & Sarvadevabhatla, R. K. (Poster at **ICDAR '26**)

[S.2] DoCoG: Mask-based Multi-Type Grounded Chain-of-Thought for Document QA
Gunda, S. M.*, Jinka, J. S.*, Rachakonda, H. S.*, Jain, A., Venna, V. K., **Srinivasan, A.** & Sarvadevabhatla, R. K. (under submission at **ECCV**)

[C.1] [W.1] [W.2] M3Grounder: Mask-Based Multi-Span and Multi-Granular Grounding for Document QA
Venna, V. K.*, Gunda, S. M.*, Jinka, J. S., Rachakonda, H. S., **Srinivasan, A.** & Sarvadevabhatla, R. K. (Poster at **CVPR'26**, MMFM5@CVPR'26 & **Oral at GRAIL-V, DataMFM @ CVPR'26**)

Work Experience

BharatGen *Hyderabad, India*

SENIOR AI RESEARCH ENGINEER MENTORED BY
DR. RAVI KIRAN SARVADEVABHATLA AND DR. GANESH RAMAKRISHNAN *Apr 2025 - Present*

- **Patram-Layout: A Foundation VLM for Reimagining Document Layout Understanding** **[S.1]**
  - Reframed layout understanding as **promptable, open-set hierarchical segmentation**, generating **layout trees + 80M masks**
  - Designed an **end-to-end VLM + segmentation model** with **hierarchy-aware grounding (HGC)** for layout parsing.
  - Trained with **LM + Dice/BCE + structural losses (containment, overlap)** on **2.5M documents / 400K+ labels**
- **Patram-Grounder: A VLM for Grounded Document Intelligence** (Presented at IndiaAI Impact Summit '26)**[C.1][S.2]**
  - Built an **end-to-end VLM + segmentation model** for DocVQA that generates **answers with interleaved grounding tokens** and **pixel-level masks** (phrase/line/block), replacing coarse bbox grounding with fine-grained segmentation
  - Introduced **multi-span, multi-granular & multi-type grounding** via **[GROUND] tokens + learned prompt embeddings**, enabling joint reasoning over **textual + graphical elements** with **hierarchical enclosure & interaction modules**
  - Scaled training on **200K–325K documents & 1.5M–2M grounded QA pairs** with **Dice+BCE + bleed suppression + hierarchy losses / DPO**, achieving **SOTA grounded DocVQA** with strong gains in both answer accuracy and grounding fidelity
- **Patram-FS: Distributed Dataset Filesystem for VLM Training**
  - Designed a transparent dataset filesystem for large-scale VLM training, packing **10M+** small image/annotation files into shard-level **TAR/JSONL** storage to eliminate FSx metadata bottlenecks and reduce S3 access from per-file GETs to shard-level fetches.
  - Enabled direct `pread()` byte-range reads over **50GB TAR image shards** and **1GB JSONL annotation shards** using unified Postgres metadata **(shard path, offset, length)**, avoiding decompression, scans, and per-shard indexes.
  - Engineered a per-node **Go daemon + Python client** over **Unix domain sockets** for shard orchestration: **5ms** batched DB lookup coalescing, cold-miss deduplication, async S3/FSx→NVMe fetches, `MADV_WILLNEED` page warming, and LRU-based NVMe eviction.
  - Optimized the hot path with stateless `open/pread/close`, OS page-cache sharing across DataLoader workers, `cv2.imdecode` + `np.frombuffer` zero-copy decode, and ENOENT self-healing, achieving ∼**0.5ms** warm image reads and ∼**2.3μs** annotation reads.
- **Patram-Core: Custom VLM Foundation Model Training Framework** **[S.1] [C.3] [S.2]**
  - Built a modular distributed training framework for **VLM foundation models**, spanning multimodal pretraining, SFT, OCR, grounding, and document understanding tasks via registry-driven model/data abstractions.
  - Engineered a scalable data engine with **60+ dataset adapters**, iterable multi-dataset mixing, task/family-specific collators, tokenizer extensions, and Hydra-configurable data mixtures.
  - Supported major model families including **QwenVL, Gemma, Molmo, InternVL, SAM-backed grounding models**, with pluggable vision encoders via **CLIP/SigLIP/MetaCLIP/OpenVision** registries.
  - Implemented large-scale training with **HuggingFace Trainer**, **DeepSpeed ZeRO-3**, `torchrun`, **bf16**, gradient checkpointing, LoRA/PEFT, selective freezing, checkpoint resume/consolidation, and multi-node launch pipelines.

- Optimized distributed evaluation with per-GPU decoding, compact metric reduction, batched `ComputeMetrics`, and **W&B** logging to avoid cross-rank validation OOMs; validated runs using NCCL `all_reduce` microbenchmarks.

- **Patram-Ingest: Scalable Data Generation Engine with LLMs and VLMs** **[S.1] [C.2] [C.3] [S.2]**
  - Built a distributed LLM/VLM data generation engine for **1B+ image-scale** inference using **Python**, **vLLM**, and **PostgreSQL/SQLite**, with atomic state transitions, deduplication, cached `state_counts`, and shard-aware JSONL outputs.
  - Optimized large-scale ingestion with batched inserts and **PostgreSQL COPY FROM STDIN**; benchmarked **12M metadata ingests** and high-throughput fetch/mark-done flows for multi-node processing.
  - Engineered crash-safe execution via a **write-fsync-commit** protocol, async JSONL writers, 1GB shard rotation, **250-sample** checkpoints, and bounded replay recovery with zero DB/output corruption.
  - Designed a stateless **bootstrap-process-flush** pipeline overlapping CPU preprocessing, prompt construction, validation/retry hooks, and GPU inference; added pluggable `PromptAndValidation` APIs for extensible multimodal generation workflows.
  - Integrated **StageWeaver**-style multi-stage orchestration with per-stage DBs, router-based DAG coordination, fan-in/fan-out routing, shared model initialization, and crash-safe cross-stage handoff via `fetch_state=4`.

- **Patram-Parse: An Agentic Document Parsing Framework** (Presented at Impact AI Summit '26)**[C.2]**
  - Built an **agentic document parsing workflow** generating **Fid-HTML** with **93% visual fidelity** outperforming Gemini 2.5 Pro.
  - Designed coordinated agents: **Segment (hierarchical layout tree)**, **Router + Specialist Agents (text/table/image/form/list/eqn)**, **Reading-Order Agent**, and **Reflection Agent**, achieving **0.97 routing accuracy / 0.96 DOM validity**
  - Achieved **SOTA 0.86 overall score** on VFDR-Bench (5K docs, 15 langs, 15 domains) with **85% physical structure fidelity**

FOUNDING ML ENGINEER MENTORED BY
DR. RAVI KIRAN SARVADEVABHATLA AND DR. GANESH RAMAKRISHNAN *Jun 2024 - Mar 2025*

- **DarshinAI: An Unified Data Platform for Annotation, Visualization and Evaluation** (Presented at BharatGen Summit '25)
  - Supports multiple Document AI tasks such as **OCR, layout, reading-order, VQA, and document conversion**
  - Designed a **scalable multi-user system** with **grounded comments, leaderboard evaluation, experiment tracking, and multi-view analysis** for large-scale document AI workflows

- **Patram-Bench: A Comprehensive Indian Document Understanding Benchmark** (Presented at BharatGen Summit '25)**[C.3]**
  - Built a **multilingual document benchmark** with **52,607 documents**, **23 languages**, **13 domains**, **71 classes** and **52 tasks**, comprising **2.4M+ QA annotations** across OCR, parsing, layout, and multi-hop DocVQA
  - Designed **Patram-Syn**: a **persona-driven synthetic pipeline** with **200K+ personas** and a **5-stage generation stack** (retrieval → structured code → multilingual rendering → human validation) for charts, tables, HTML and diagrams
  - Introduced **multilingual robustness metrics** (**DUCLI, SRS**) and evaluated **29+ VLMs**, exposing **cross-lingual degradation**, **script sensitivity**, and failures in **layout-intensive, low-resource settings**

- **Patram-DB: High-Performance Multimodal Database for VLM Pipelines**
  - Built a PostgreSQL + FastAPI unified data platform for large-scale VLM training, managing **50M+ document images** and **250M+ multimodal annotations** via a sharded architecture with declarative partitioning.
  - Engineered a high-concurrency ingestion pipeline achieving **65k+ rows/sec** peak throughput using async multi-threaded PostgreSQL `COPY`, `UNLOGGED` staging tables, and batched transactions.
  - Achieved <**1.5 ms P99 query latency** by implementing PgBouncer connection pooling, query partition pruning, and JSONB with GIN/B-tree composite covering indexes for open-set metadata retrieval.
  - Guaranteed real-time referential consistency and audit-trailed versioning across distributed storage using C-level triggers, materialized views, and TOAST management to prevent bloat from massive OCR payloads.

- **Patram: India's First VLM for Document Understanding** (Launched at BharatGen Summit '25)
  - Scraped **BharatDocs**, the largest available Indian document corpus, with **20M+ documents** and **250M+ pages** across **English + 22 Indian languages**.
  - Pre-trained on **1M+ images** for captioning, followed by **500K VQA annotations** across **20+ document VQA task types**.
  - Built a multi-stage **Synthetic QA pipeline** using **Qwen2.5-VL-72B-Instruct** for diverse QA generation across a taxonomy of extractive and abstractive QA types, followed by **verification** and **deduplication** stages.
  - Achieved state-of-the-art (SoTA) performance on key DocVQA benchmarks, including DocVQA and VisualMRC

- **e-vikrAI: Indic VLMs for e-commerce** (Presented at IMC '24)
  - Built an **e-commerce VLM** for auto-cataloging, generating **SEO-optimized titles**, hierarchical categories, descriptions, and metadata **(price, size, color)** directly from product images.
  - Built a scalable e-commerce scraper for **22.8M+ products**, covering category/product/metadata/image pipelines with **hierarchical link extraction**, anti-bot/CAPTCHA handling, rotating proxies, retry/backoff, checkpointed resume, and high-throughput ingestion via **async I/O, multiprocessing, and Scrapy/Twisted**.
  - Developed a PoC mobile app demonstrating end-to-end cataloging with **multilingual translation** and **text-to-speech (TTS)** for improved accessibility.

- Reduced cataloging time from **10 min/product** manually to **6 sec/product**, enabling scalable seller onboarding and simplified usage for **rural/non-English-speaking users**.

Pertinent Skills

Programming	Python, CUDA, C++, JavaScript, DBs (faiss, mongodb, postgresql)
ML / VLM Stack	PyTorch, OpenCV, HF transformers, HF datasets, webdataset, TRL, deepspeed, accelerate, NCCL, slurm
Optimization	triton, bitsandbytes, flash-attn, xformers, vLLM, sglang, ollama, cuML, cuDF, ONNX, TensorRT
Tools	git, linux, bash, wandb, gradio, streamlit, nsys, ncu, docker, fastapi

Selected Scholastic Achievements

2024	**Institute Gold Medal** for Excellence in Academics and Co-curricular Activities (Rank 1 out of 1103), IIT Hyderabad
2024	**Gold Medalist**, IEEE Signal Processing Cup, *ICASSP '24, South Korea*
2023	**Silver Medalist**, ISRO Mid-Prep Event, *Inter IIT Tech Meet 11.0, IIT Kanpur*
2022	**Silver Medalist**, Bosch Mid-Prep Event, *Inter IIT Tech Meet 10.0, IIT Kharagpur*
2021	**Academic Excellence Awardee**, *13th Foundation Day of IIT Hyderabad*
2020	**100.0000000 percentile** in JEE Mains Chemistry
2019	**All India Rank 333**, Kishore Vaigyanik Protsahan Yojana (KVPY) Scholar, SA Stream

Volunteering and Community Service

2026	**Reviewer @ ICDAR '26**, for submissions pertaining to Document Layout Analysis & Document Reconstruction
2025	**Reviewer @ AAAI '26**, for submissions pertaining to VLM Benchmarks
Fall '23	**Head Teaching Assistant @ Dept. of CSE, IIT-H**, Data Structures lectured by Dr. Maria Francis & Dr. M.V. Panduranga Rao
Fall '22	**Teaching Assistant @ Dept. of CSE, IIT-H**, Intro to Programming lectured by Dr. Karteek S & Dr. Ramakrishna Upadrasta

Exploratory Undergrad Research

- **IEEE Signal Processing Cup 2024 (ICASSP) – ROBOVOX**: Secured 1st place globally; built a far-field speaker recognition system under real-world noise conditions (ambient/internal noise, reverberation, distance, babble, angle variability); developed baseline via score fusion of SEResNet and ECAPA-TDNN, and improved with ERes2Net + augmentations (RIR, MUSAN, speed) and adaptive s-norm scoring; achieved minDCF 0.67, EER 8.93% (vs baseline: 0.7096, 9.59%) under Dr. Sri Rama Murty Kodukula
- **Quantum Methods for Vehicle Routing Problem (VRP)**: Surveyed classical (LP via CPLEX) and quantum (CQM via D-Wave) approaches with focus on Multiple Vehicle Visits (MVV); implemented spherical q-means in Qiskit for Multi-Depot VRP (MD-VRP), achieving DBI 0.742 and improved routing cost across 13 datasets; demonstrated superior CQM runtime and explored QUBO formulations under Dr. M.V. Panduranga Rao
- **LiDAR Semantic Segmentation for Autonomous Driving**: Built an end-to-end pipeline; enhanced RangeSeg with bicubic and Lanczos interpolation, improving performance on SemanticKITTI, SemanticPOSS, nuScenes; proposed SRGAN-based upsampling for challenging scenarios under Dr. C. Krishna Mohan
- **Neuro-Fuzzy Clustering for Gaussian Mixture Models (GMM)**: Implemented ANN-based reformulation of Expectation Maximization and developed Neuro-Fuzzy C-Means in Python using DEAP; achieved +11.8% improvement in Silhouette Score over GMM under Dr. Kishalay Mitra
- **Adversarial Robustness for IR2Vec**: Performed adversarial training to mitigate targeted attacks; generated adversarial examples using FGSM and explored Discrete Adversarial Manipulation of Programs (DAMP) under Dr. Ramakrishna Upadrasta
- **Neuromorphic AI Systems Survey**: Surveyed biologically inspired models from Hebbian Learning and MP Neuron to Hopfield Networks, Spiking Neural Networks, and Cellular Neural Networks, bridging neuroscience and modern ML under Dr. Mohan Raghavan
- **MotionGPT for Yoga Postures**: Extended MotionGPT; built MoCap pipelines (C3D, AMASS, SMPL, ASF, AMC); trained on in-house data under Dr. Mohan Raghavan

Positions Of Responsibility and Extracurriculars

- **Mentor, Epoch (AI-ML & Data Science Club)** @ IIT Hyderabad *('23-'24)*
- **Overall Head, Epoch** @ IIT Hyderabad *('22-'23)*
- **Deputy Contingent Leader** @ Inter IIT Tech Meet 11.0 *('22-'23)*
- **ML Lead, Google Developers Student Club** @ IIT Hyderabad *('22-'23)*
- **Student Head, Sunshine UG Mentorship Program** @ IIT Hyderabad *('22-'23)*
- **Core Member, Epoch** @ IIT Hyderabad *('21-'22)*
- **Core Member, Robotix** @ IIT Hyderabad *('21-'22)*
- **Campus Tour Coordinator** @ Alumni Cell, IIT Hyderabad *('23-'24)*
- **DUGC Student Member** @ CSE, IIT Hyderabad *('21-'24)*
- **Class Representative** @ CSE, IIT Hyderabad *('21-'22)*
- **Social Media Manager** @ CSE, IIT Hyderabad *('21-'24)*
- **UG Mentor** @ Sunshine, IIT Hyderabad *('21-'22)*
- **Workshop Organizer, Ethical Hacking** @ Elan & Nvision, IIT Hyderabad *('20-'21)*
- **Silver Medal, ISRO Mid-Prep** @ Inter IIT Tech Meet *('23)*
- **Silver Medal, Bosch Mid-Prep** @ Inter IIT Tech Meet *('22)*
- **Silver Medal, Freestyle Duo (Whistle + Beatbox)** @ Milan *('23)*
- **Gold Medal, Short Film Competition** @ Milan *('22)*
- **Gold Medal, Institute Cricket League** @ IIT Hyderabad *('21)*
- **Bronze Medal, Cricket** @ Diesta *('21)*
- **Silver Medal, Table Tennis & Tennis** @ Diesta *('21)*
- **Silver Medal, Cricket & Basketball** @ Milan *('22-'23)*
- **Bronze Medal, Swimming** @ Milan *('22)*